

Arrhenius Reactions

GEM allows use of Arrhenius type rate reactions in aqueous phase. The rate of concentration (molality) change using Arrhenius rate reaction is given by,

$$r = F \cdot \exp \left[- \frac{E_a}{R} \left(\frac{1}{T} - \frac{1}{T^*} \right) \right] \prod_{i=1}^{\text{no. of reactants}} c_i^{a_i} \quad (\text{D9.1})$$

for a reaction with (no. of reactants), where, E_a is activation energy [J/mol] and R is the gas constant [8.314 J/ mol K]. The concentration of reactants c_i is in molality [mol/kg H₂O] and the order of reactions for the reactants is given by a_i . F is frequency factor or pre-exponential factor and its unit depends on the overall order of the reaction, i.e. [1/(sec.molality^{a-1})] where a is overall order of reaction [1/(sec.molality^{a-1})] where a is overall order of

reaction = $\sum_i^{\text{no. of reactants}} a_i$. Temperature T is expressed in [K].